Reflecitve Report:

Evaluating Apple's MLX Framework for Machine Learning: A Comparative Study Using Unsupervised Clustering Algorithms

Professor: **Prof. Vahab Esfandani**

M513B Research Seminar

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Introduction

This reflective report analyzes the design and execution of the research project *"Evaluating Apple's MLX Framework for Machine Learning: A Comparative Study Using Unsupervised Clustering Algorithms"*. The project investigated how MLX performs for KMeans clustering compared with PyTorch, NumPy, and scikit-learn on Apple Silicon hardware, using both the MNIST and a credit card customer dataset. As a second-year MEng Computer Science student with a background in software engineering and practical experience with machine learning libraries and Apple's development ecosystem, this topic aligned closely with existing skills and long-term interests. The reflection focuses on personal learning in three areas: research design, technical experimentation with MLX, and the interpretation and communication of empirical findings.

Personal and Peer Context

Before undertaking this research project, prior experience with PyTorch, NumPy, and scikit-learn, as well as Apple's frameworks on macOS and iOS, was established, which meant that fundamental machine learning concepts and tools were already familiar. However, formal research design, systematic benchmarking, and critical reflection on methods had been more limited. A strong personal motivation was a bullish view on on-device machine learning and a desire to understand whether MLX can realistically support classical algorithms for edge scenarios, not just large language models or transformer workloads.

Peer interaction during the module also partially shaped the project. Early discussions in class helped narrow the focus from a broad comparison of frameworks on Apple Silicon to a clearer question about MLX and KMeans clustering. These exchanges pushed the project away from a purely technical benchmark towards a more critical reflection on problem framing.

Designing the Research

Designing the research proposal was an important learning phase. The assessment brief required an analytical reflection on research design, including identification of key learning moments and their significance. Translating this requirement into practice meant explicitly articulating the research gap, structuring research questions, and connecting them to a feasible methodology.

The literature review showed that current work on MLX focuses mainly on deep learning and transformers, with little coverage of traditional algorithms such as KMeans. Recognising this gap helped to formulate specific questions about computational performance, clustering quality, API usability, and practical recommendations for on-device ML. Adopting a framework-comparison style methodology, inspired by prior work on benchmarking machine learning libraries, required thinking carefully about experimental controls, reproducibility, and validity. This process improved understanding of how theoretical concepts in research methods translate into concrete design decisions such as fixed random seeds, consistent preprocessing, and unified evaluation metrics.

Deciding to treat the credit card dataset as a case study for customer churn highlighted the importance of aligning methods with business questions. Initially, there was an implicit assumption that KMeans clusters would map meaningfully onto churn labels. Refocusing the study to treat this as a hypothesis rather than a given reinforced the need to keep research questions open and testable rather than confirmatory.

Conducting the Experiments

Implementing KMeans across MLX, PyTorch, NumPy and scikit-learn provided several technical learning opportunities. The goal was to keep algorithmic logic constant while varying the underlying framework so that differences in performance and memory usage could be attributed to implementation rather than design. In practice, this ideal proved challenging, and grappling with those challenges deepened appreciation for the subtleties of empirical work.

MLX posed specific implementation issues. The absence of Boolean indexing and an exact equivalent to NumPy's random-choice routines required mask-based workarounds and permutation-based initialisation strategies. These constraints forced a more deliberate approach to array operations and made the impact of lazy evaluation and type systems more visible. Working through these issues improved confidence in reading and reasoning about unfamiliar APIs and underscored the importance of careful debugging and incremental testing when dealing with emerging frameworks.

The results for the credit card dataset showed that all frameworks achieved very similar clustering quality: Adjusted Rand Index was close to zero and silhouette scores were around 0.52, indicating that while clusters were internally coherent, they did not align with the hidden churn labels. At the same time, MLX delivered competitive runtime and lower memory usage compared with scikit-learn and NumPy, especially in terms of post-fit memory consumption. Observing these outcomes challenged initial expectations that better hardware-aware frameworks would automatically yield more useful business insight. Instead, the experiments highlighted that algorithm choice and feature engineering matter at least as much as framework efficiency.

Learning from Interpretation and Feedback

Interpreting the findings and turning them into a coherent narrative for the presentation and this report was another key learning moment. The assessment brief emphasises that reflective essays must move beyond description to analysis, supported by evidence from the course. In practice, this meant asking what the results reveal about both MLX and personal practice as a researcher.

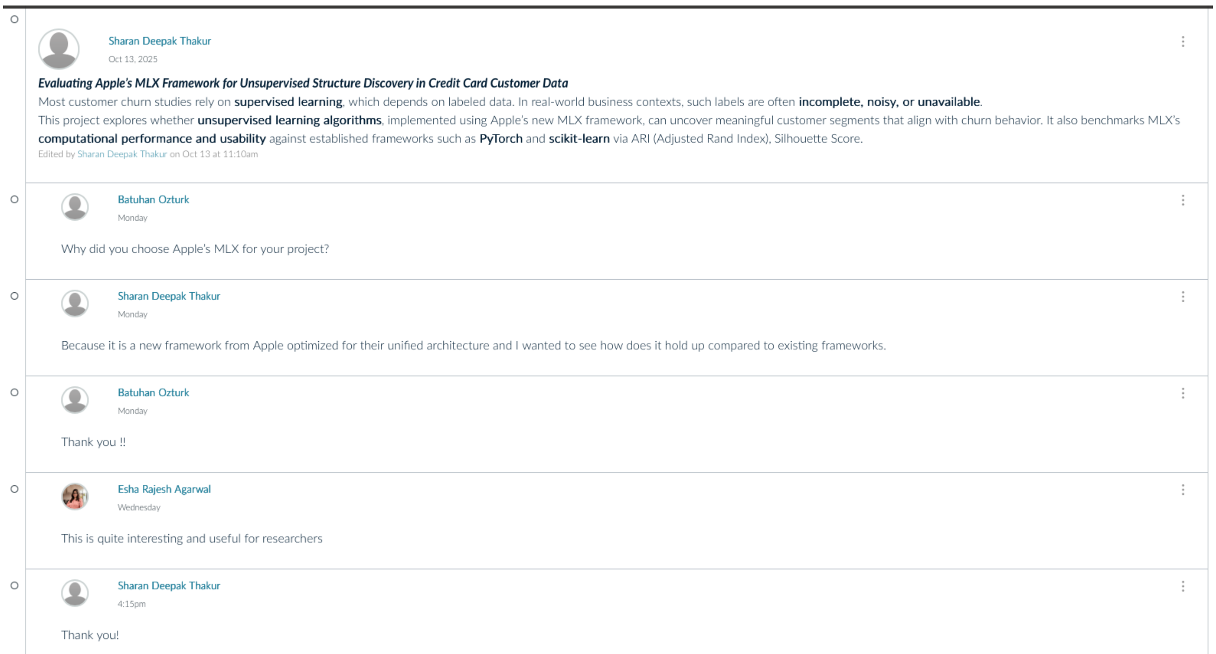
A major insight was the distinction between *technical* and *substantive* success. Technically, the project demonstrated that MLX is a viable and memory-efficient option for classical KMeans on Apple Silicon, with performance that is competitive for small to medium datasets. Substantively, the failure of clustering to predict churn showed that even a well-executed experiment can deliver negative or inconclusive results regarding the original business question. Learning to accept and still value such results as meaningful contributions was a significant shift in mindset.

Instructor feedback on the presentation pointed out that the methodology and conceptual framework were presented together in a way that obscured their distinct roles. Acting on this feedback meant creating separate, simpler figures with clearer boundaries between theoretical motivation and empirical procedure. This revision process reinforced an important lesson: effective research communication requires not just technical accuracy but also careful attention to how information is structured and layered for different audiences.

Conclusion

This research and reflective process have increased my awareness of both the technical and conceptual dimensions of machine learning experimentation on new hardware platforms. I have gained practical skills in benchmarking, troubleshooting, and critical analysis, while learning the value of adaptation and iterative design. The experience has reinforced my interest in bridging academic research with real-world ML development and has prepared me to pursue further work in evaluating lightweight frameworks for edge computing and on-device ML.

Screenshots of Discussion Forum



Mohit Harpalani
Oct 13, 2025

The Death of Privacy: AI Agents That Learn and Mimic Human Behavior

Idea:
Investigate how advanced AI learns personal habits, tone, and behavior — and how this erodes digital privacy, trust, and even individuality.
Focus:

- Behavioural cloning and psychological profiling
- Human–AI indistinguishability risks
- Ethical frameworks for digital mimicry

OR

"Impact of AI Robots on Employment: Opportunities, Threats, and Workforce Adaptation"

Rohan Mathew
Oct 27, 2025

AI on employment sounds more practical !

Sharan Deepak Thakur
Nov 3, 2025

How do you plan to answer your research question?

Reply

Batuhan Ozturk
Oct 13, 2025

Evaluating GNU Radio for Noise Performance Analysis in Software-Defined FM Reception

Traditional FM receivers rely on physical hardware to process and decode signals.
In contrast, GNU Radio, an open-source digital signal processing (DSP) framework, enables these operations entirely in software — creating a flexible software-defined radio (SDR) system.

This project builds a software-based FM receiver using GNU Radio and explores how different Signal-to-Noise Ratio (SNR) levels affect signal quality and audio clarity.
It applies controlled noise, analyzes performance, and visualizes the results to understand how software processing can replicate and optimize traditional radio behavior.

The innovation lies in demonstrating how SDR and GNU Radio can be used as a low-cost, adaptable platform for real-world applications, including IoT and Smart Energy systems, where efficient and scalable wireless data communication is essential.

Sharan Deepak Thakur
Nov 3, 2025

What are the research questions?

Batuhan Ozturk
Monday

How does noise influence the reliability of FM signal reception?

To what extent can software-defined radio replicate the performance of traditional hardware radios?

How practical is an SDR-based system for real-world IoT and smart-energy data transmission?

Sharan Deepak Thakur
Monday

Thanks!

Reply

Reply

Venkata Skandha Rajendra Varma
Oct 28, 2025

My topic is **Adaptive Spiking Neural Networks for Ultra-Low-Latency High-Frequency Trading -**

This research explores the use of Adaptive **Spiking Neural Networks (SNNs)** to achieve **ultra-low-latency decision-making** in **high-frequency trading (HFT).** By encoding order-book and volume data as spike trains, the model reacts to market micro-events in real time, offering an event-driven alternative to traditional ML models. The study investigates whether SNNs can provide faster and more energy-efficient trade signal generation while maintaining competitive predictive accuracy.

Sharan Deepak Thakur
Nov 3, 2025

Will this be a comparative study?

Afinan Karunnappilly Naushad
Nov 28, 2025

Are you planning to compare your results with existing industry standards?

Reply

Afinan Karunnappilly Naushad
Oct 28, 2025

AI-Powered Predictive Analytics for Risk and Safety Management in Construction Projects-

Construction projects are a complicated matter by themselves, and any safety risks and possible delays will affect the well-being of workers as well as the cost of the projects. The traditional risk management approach is